DTFT and its properties

Answers and worked solutions · Day 06 · Thu 29 Oct · pack D3

5. Practice

From your course files

T4 Q1 (tutorial).

(a) Finite geometric series with 6 terms (compare with the figure in Section 1):

$$X(\omega)=\sum_{k=0}^{5}e^{-j\omega k}=\frac{\mathbf{1-e^{-j6\omega}}}{\mathbf{1-e^{-j\omega}}}\qquad\left(=e^{-j5\omega/2}\frac{\sin 3\omega}{\sin(\omega/2)}\right)$$

(b) Substitute $n=-k$; the ratio is $\frac{1}{2}e^{j\omega}$:

$$X(\omega)=\sum_{k=-\infty}^{0}2^{k}e^{-j\omega k}=\sum_{n=0}^{\infty}\left(\tfrac{1}{2}e^{j\omega}\right)^{n}=\frac{\mathbf{1}}{\mathbf{1-\frac{1}{2}e^{j\omega}}}$$

(c) Change variable $n=k+4$; the constant is $3^4e^{j4\omega}$:

$$X(\omega)=\sum_{k=-4}^{\infty}\left(\tfrac{1}{3}\right)^{k}e^{-j\omega k}=3^4e^{j4\omega}\sum_{n=0}^{\infty}\left(\tfrac{1}{3}e^{-j\omega}\right)^{n}=\frac{\mathbf{81e^{j4\omega}}}{\mathbf{1-\frac{1}{3}e^{-j\omega}}}$$

(d) The ↑ is under the 0. Pair the terms and use Euler's identity:

$$X(\omega)=-2e^{j2\omega}-e^{j\omega}+e^{-j\omega}+2e^{-j2\omega}=\mathbf{-2j\big(2\sin(2\omega)+\sin(\omega)\big)}$$

Real and odd x gives purely imaginary, odd X.

T4 Q2 (tutorial).

The ↑ is under the -3, so $x(0)=-3$. Use the section 1.4 shortcuts only.

- a. $X(0)=\sum_k x(k)=-1+2-3+2-1=\mathbf{-1}$
- b. $\angle X(0)=\angle e^{\pm j\pi}=\pm\pi$ (a negative real number, not 0)
- c. $\int_{-\pi}^{\pi}X(\omega)\,d\omega=2\pi x(0)=\mathbf{-6\pi}$
- d. $X(\pi)=\sum_k x(k)e^{-j\pi k}=\sum_k x(k)(-1)^k=-1-2-3-2-1=\mathbf{-9}$
- e. $\int_{-\pi}^{\pi}|X(\omega)|^2d\omega=2\pi\sum_k|x(k)|^2=2\pi(1+4+9+4+1)=\mathbf{38\pi}$

T4 Q3 (tutorial).

(a) $H(\omega)=\dfrac{Y(\omega)}{X(\omega)}=\dfrac{\mathbf{1-\frac{1}{2}e^{-j\omega}}}{\mathbf{1-\frac{1}{8}e^{-j\omega}}}$

(b) $e^{j\omega n}$ is an eigenfunction of LTI systems, so $\mathbf{H(\pi/5) = 3e^{-j\pi/3}}$. It is not possible to determine $H(\omega)$ for other values of ω from this one pair.

(c) $u(n) - u(n-1) = \delta(n)$, so $y = h$: $h(n) = \left(\frac{1}{5}\right)^{n+2} u(n+2)$. That is $\left(\frac{1}{5}\right)^{n} u(n)$ shifted by -2, so $\mathbf{H(\omega) = \dfrac{e^{j2\omega}}{1 - \frac{1}{5}e^{-j\omega}}}$.

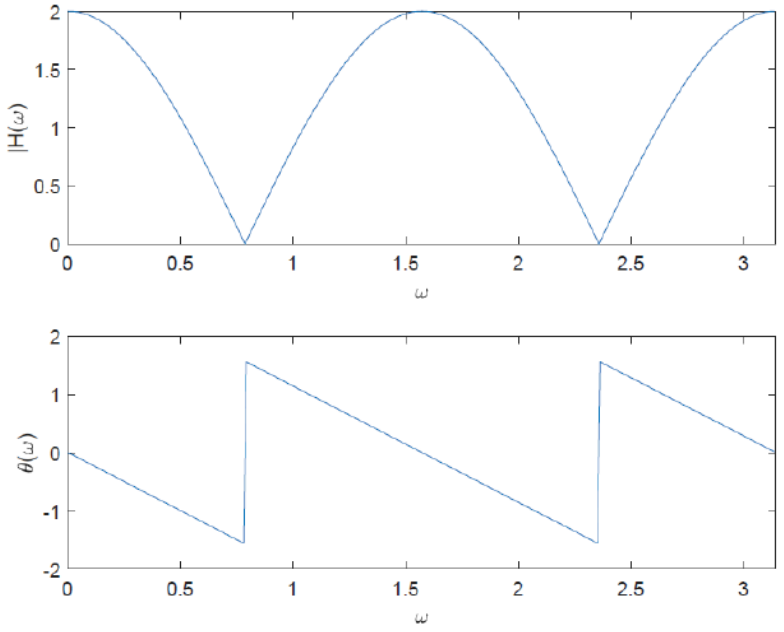
T4 Q4 (tutorial).

(a) Time-shift property:

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = 1 + e^{-j4\omega} = e^{-j2\omega}(e^{j2\omega} + e^{-j2\omega}) = 2e^{-j2\omega}\cos(2\omega).$$

$$|H(\omega)| = 2|\cos(2\omega)|, \qquad \theta(\omega) = \begin{cases} -2\omega, & \cos(2\omega) \geq 0 \\ -2\omega + \pi, & \text{otherwise} \end{cases}$$

(π is added to absorb the multiplication by -1 into the exponent.)



T4 Q4(a): the official magnitude and phase sketches (from the tutorial solutions)

(b)(i) $H(\pi/4) = 0$ removes the first tone. $H(\pi/3) = 2e^{-j2\pi/3}\cos(2\pi/3) = e^{j\pi/3}$, so

$$y(n) = \mathbf{3}\sin\left(\tfrac{\pi}{3}\mathbf{n} + \tfrac{\mathbf{13\pi}}{\mathbf{30}}\right)$$

(ii) $H(0) = 2$. $|H(2\pi/5)| = 1.6180$ and $\theta(2\pi/5) = -4\pi/5 + \pi = \pi/5$, so

$$y(n) = 20 + 5(1.6180)\cos\left(\tfrac{2\pi}{5}n + \tfrac{\pi}{2} + \tfrac{\pi}{5}\right) = \mathbf{20 + 8.0902}\cos\left(\tfrac{\mathbf{2\pi}}{\mathbf{5}}\mathbf{n} + \tfrac{\mathbf{7\pi}}{\mathbf{10}}\right)$$

The full method is in D4 (Example 1).

New questions

N1 [3 marks] (easy).

a. $X(\omega) = e^{j\omega} + 1 + e^{-j\omega} = \mathbf{1 + 2}\cos\omega$. It is real because $x(n)$ is real and even.

b. $X(0) = 3 = \sum x$. $X(\pi) = 1 - 2 = \mathbf{-1} = \sum(-1)^n x(n) = -1 + 1 - 1$. (checks)

N2 [4 marks].

a. The sum starts at $n = -1$. Let $m = n + 1$:

$$X(\omega) = \sum_{m=0}^{\infty}(-0.5)^{m-1}e^{-j\omega(m-1)} = (-0.5)^{-1}e^{j\omega}\sum_{m=0}^{\infty}(-0.5e^{-j\omega})^m = \boxed{\frac{-2e^{j\omega}}{1 + 0.5e^{-j\omega}}}$$

b. $X(0) = \sum_{n\geq -1}(-0.5)^n = -2 + \frac{1}{1.5} = \mathbf{-4/3}$. From (a): $-2/1.5 = -4/3$ (checks). $X(\pi) = \sum(-0.5)^n(-1)^n = \sum_{n\geq -1} 0.5^n = 2 + 2 = \mathbf{4}$. From (a): $-2(-1)/(1 - 0.5) = 4$ (checks).

c. $|X(\pi)| = 4 > |X(0)| = 1.33$, so the energy is concentrated near $\omega = \pi$. It is **highpass**. The alternating sign $(-1)^n$ moves a lowpass 0.5^n spectrum by π (modulation property).

N3 [5 marks].

Indices: $x(-2) = 2,\ x(-1) = -1,\ x(0) = 3,\ x(1) = 0,\ x(2) = 1$.

a. $X(0) = 2 - 1 + 3 + 0 + 1 = \mathbf{5}$

b. $X(\pi) = \sum(-1)^n x(n) = 2(+1) + (-1)(-1) + 3(+1) + 0 + 1(+1) = \mathbf{7}$

c. $2\pi x(0) = \mathbf{6\pi}$

d. $2\pi(4 + 1 + 9 + 0 + 1) = \mathbf{30\pi}$

e. The inverse DTFT at $n = 2$ gives $\int Xe^{j2\omega}d\omega = 2\pi x(2) = \mathbf{2\pi}$. (Using $x(-2)$ by mistake gives 4π.)

f. $dX/d\omega = \sum(-jn)x(n)e^{-j\omega n}$, so at $\omega = 0$ it equals $-j\sum nx(n) = -j[(-2)(2) + (-1)(-1) + 0 + 0 + 2(1)] = -j(-1) = \mathbf{j}$.

All six were checked by numerical integration and differentiation.

N4 [6 marks] (multi-part, chained).

a. Convolution property: $Y(\omega) = X(\omega)^2 = \boxed{\dfrac{1}{(1-0.5e^{-j\omega})^2}}$

b. From Example 3(a), $n(0.5)^n u(n) \leftrightarrow \dfrac{0.5e^{-j\omega}}{(1-0.5e^{-j\omega})^2}$. Add $(0.5)^n u(n) \leftrightarrow \dfrac{1-0.5e^{-j\omega}}{(1-0.5e^{-j\omega})^2}$:

$$\frac{0.5e^{-j\omega} + 1 - 0.5e^{-j\omega}}{(1-0.5e^{-j\omega})^2} = \frac{1}{(1-0.5e^{-j\omega})^2} = Y(\omega)$$

So $y(n) = (n+1)(0.5)^n u(n)$. (A direct numerical convolution gives 1, 1, 0.75, 0.5, 0.3125, ... (checks))

c. $Y(0) = 1/0.5^2 = \mathbf{4}$. Check: $\sum (n+1)0.5^n = 4$ (checks). $Y(\pi) = 1/1.5^2 = \mathbf{0.444}$.

d. $V(\omega) = Y(2\omega) = \boxed{\dfrac{1}{(1-0.5e^{-j2\omega})^2}}$. $V(0) = Y(0) = \mathbf{4}$ (inserting zeros does not change the sum). $V(\pi) = Y(2\pi) = Y(0) = \mathbf{4}$: this is the image created by upsampling.

N5 [5 marks] (sampling).

a. 150 Hz: $\omega = 2\pi(150)/500 = \mathbf{0.6}\pi$. 400 Hz: $\omega = 1.6\pi$. Subtract 2π to get $\mathbf{-0.4}\pi$, so its lines are at $\pm 0.4\pi$. Apparent frequency $= 0.4\pi \cdot 500/(2\pi) = 100$ Hz. **It aliases.**

b. Impulses of area π at $\omega = \pm 0.6\pi$ and of area 0.5π at $\pm 0.4\pi$, repeated every 2π.

c. $\cos(\mathbf{2\pi\,150t}) + \mathbf{0.5}\cos(\mathbf{2\pi\,100t})$

d. $F_s > 2(400) = \mathbf{800\ Hz}$

N6 [5 marks] (hard: modulation and windowing).

a. $\cos(\pi n) = \frac{1}{2}(e^{j\pi n} + e^{-j\pi n})$, so $Y(\omega) = \frac{1}{2}[X(\omega-\pi) + X(\omega+\pi)]$. Since X has period 2π, $X(\omega+\pi) = X(\omega-\pi)$, which gives $\boxed{Y(\omega) = X(\omega-\pi) = \dfrac{1-e^{-j6\omega}}{1+e^{-j\omega}}}$ (because $e^{-j6(\omega-\pi)} = e^{-j6\omega}$). Equivalently, $\cos(\pi n) = (-1)^n$.

b. $Y(0) = X(-\pi) = \sum (-1)^n x(n) = 1-1+1-1+1-1 = \mathbf{0}$.
$Y(\pi) = X(0) = \mathbf{6}$.

c. $|y(n)|^2 = 1$ for 6 samples, so the integral is $2\pi \cdot 6 = \mathbf{12}\pi$. Modulation does not change the energy.

d. It is the figure's curve moved by π: the main lobe of height 6 is now centred at $\omega = \pm\pi$ and there is a zero at $\omega = 0$. The lowpass pulse has become a highpass signal.

6. Quick recall check

Question

1. $4 - 1 + 2 = 5$.
2. $x(-n-3) = x(-(n+3))$: reverse, then advance by 3, which gives $e^{j3\omega}X(-\omega)$.
3. $\omega = 2\pi(1200)/2000 = 1.2\pi$, which reduces to -0.8π. The apparent frequency is $0.8\pi \cdot 2000/(2\pi) = 800$ Hz.
4. The analysis equation: $c_k = \frac{1}{N}\sum x(n)e^{-jk\omega_0 n}$. c_0 is the mean (dc) value over one period.
5. $X(\omega)$ is purely imaginary and odd, e.g. T4 Q1(d), $-2j(2\sin 2\omega + \sin\omega)$.